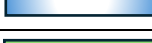
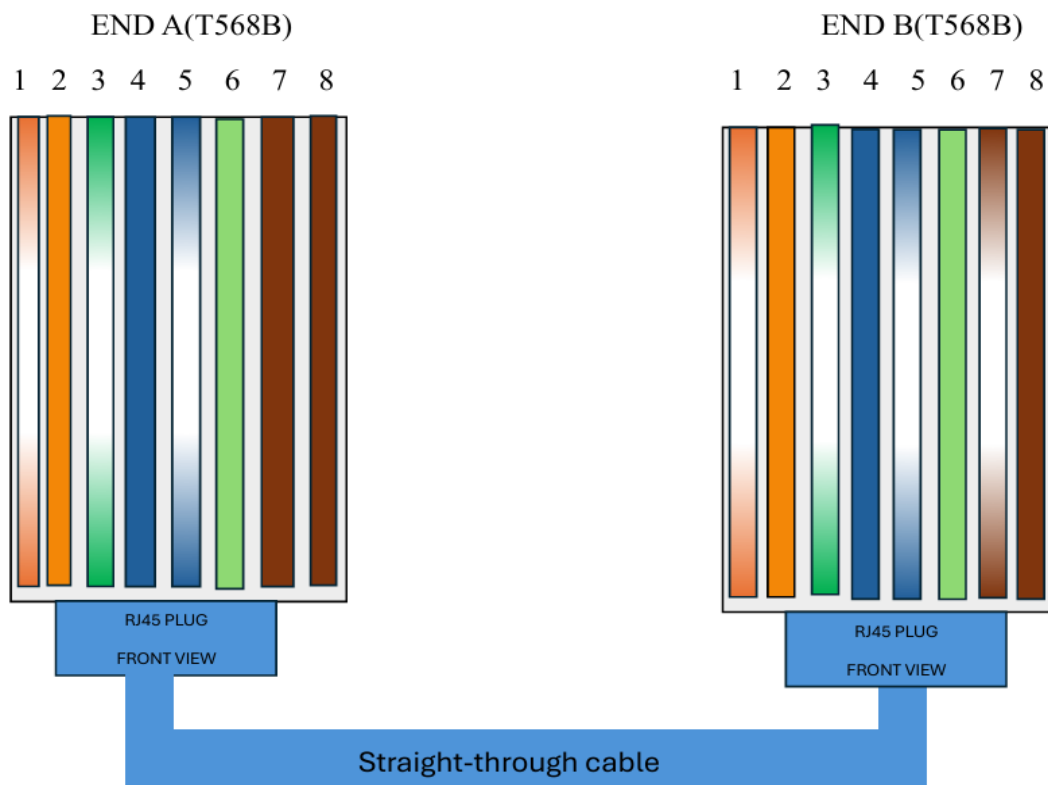


Network cable and connectors

1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Pin END(A)	Wire color	Pin END(B)
1	 White/orange	1
2	 Orange	2
3	 White/green	3
4	 Blue	4
5	 White/blue	5
6	 Green	6
7	 White/brown	7
8	 Brown	8



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Network Cable Type	Real-World Device/Application	Why It Is Used
Cat5e Cable	WiFi Router	Used to connect a router to a computer or modem. It supports speeds up to 1 Gbps, making it suitable for home and small office networks.
Cat6 Cable	Zomato POS (Point of Sale) Terminal	Used in restaurants to connect the POS terminal to the network. It supports speeds up to 10 Gbps (up to 55 m), providing faster and more reliable data transfer for order processing and billing.
Fiber Optic Cable	IPL Streaming Server / Data Center	Used to connect high-speed servers because it provides very high bandwidth, long-distance communication, and is immune to electromagnetic interference, making it ideal for live video streaming with minimal delay.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Easy Steps to Crimp an RJ-45 Connector

1. Cut the network cable.
2. Remove a small part of the outer cover.
3. Straighten the 8 wires.
4. Arrange the wires in the **T568B** color order.

5. Cut the wires to the same length.
6. Insert the wires into the RJ-45 connector.
7. Use a crimping tool to press the connector.
8. Test the cable with a cable tester.

Two Common Mistakes to Avoid

1. Wrong wire color order – The cable will not work.
2. Wires not fully inserted – The connection may fail.

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cable/Connector	Used For	Reason
Cat6 Ethernet Cable	Connecting gaming consoles and PCs to a network switch/router	Provides high-speed (up to 10 Gbps) and stable internet with low lag.
RJ-45 Connector	Attached to both ends of the Cat6 cable	Standard connector used to plug the Ethernet cable into consoles, PCs, switches, and routers.
HDMI Cable	Connecting consoles to TVs or projectors	Delivers high-quality HD video and audio for smooth IPL streaming and gaming.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.
Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



- Cable/Connector Type: RJ-45 Ethernet cable (likely Cat5e or Cat6).
- Devices Connected:
 - The grey cable is connected to the WAN (Internet) port (blue port), which connects the router to the modem or Internet Service Provider (ISP).
 - The yellow cable is connected to one of the LAN (Local Area Network) ports (orange ports), which connects the router to a device such as a desktop computer, laptop, network switch, or smart TV.